

Evidence boundaries in transferring biological computation to artificial systems

NeuroConverge internal manuscript candidate — 3 October 2026. This version adds a source-informed synthetic V1 figure/ground test to the retrospective portfolio and earlier M2/M5/DMP tests. It is not submission-ready: project-level audits classify both NMI Article and Analysis as not ready; source freeze, licensing, attribution, author information and venue fit remain open.

Abstract

Biological provenance does not itself establish artificial utility. We synthesize two heterogeneous research portfolios and report four separately analysed synthetic tests. The tested connectome assay and single-specimen Fish1.5 analysis did not establish a general structural advantage; its frozen topology-null inference was invalid. Published *C. elegans* evidence supports a bounded motor-state computation, whereas a planned animal-level reanalysis was schema-blocked. Realized-state input did not improve synthetic action-conditioned tracking; temporal correspondence helped a local rule but generic controls showed larger effects; and no DMP task met its mechanism-specificity gate. A source-informed synthetic V1 figure/ground test produced an interaction opposite to its stated alignment-benefit prediction, while the aligned feature-selective circuit underperformed a nonselective pool. None of these results validates biological mechanisms or general transfer. They motivate an evidence framework that separates source support, information correspondence and artificial utility, but do not validate the framework itself.

Biological systems are increasingly used to motivate artificial architectures, representations and learning rules. Yet evidence that a circuit has a particular structure or supports a biological computation does not specify which signal, representation, objective or learning procedure an artificial system should use. Connectome-derived reservoirs have been used to study task-specific computational capacity [1, 2], and recent work has transferred synaptic-diversity concepts into artificial networks and evaluated their benchmark performance [11]. Bidirectional comparison of macaque neural activity and artificial-network features has also shown that forward neural predictivity need not imply that model units are recoverable from biological activity [12]. These precedents show that biology-to-AI translation is established; they also make clear that each claim depends on the transferred variable, task and comparator. Figure 1 frames this motivation: biological provenance can guide a model, but does not by itself establish utility.

Three claims are often conflated: whether a biological structure is reliably characterized, whether it supports a particular computation, and whether an abstraction improves an artificial task. Evidence for one claim does not settle the next. For the portfolio map, we separate artificial transfer into two questions—whether task-relevant information is represented and whether that representation improves utility—yielding four organizing questions: biological structure, source-supported computation, information correspondence and artificial utility. These are distinct evidence categories, not a causal sequence or necessary conditions already established by these data.

NeuroMotif and NeuroMech approached different portions of this chain. NeuroMotif asked whether connectome-derived organization related to selected dynamical properties and whether a broader structure-to-AI claim could be supported. NeuroMech examined named biological computations, including motor-state feedback and cerebellar temporal learning, and tested simplified implementations under synthetic tasks. These histories are heterogeneous: they include source literature, source-data analyses, model reproductions, synthetic experiments, negative results, invalid analyses and blocked data paths.

Here we ask: **what can this project portfolio establish about the transfer of biologically grounded computation into artificial systems?** We treat this as a retrospective evidence audit, not as a pooled test or a claim that the projects were prospectively designed as one study. Figure 2 uses the same four questions—biological structure, source-supported computation, information correspondence and artificial utility—to show where each portfolio contributes evidence and where transitions require separate tests. Our contribution is an evidence-bounded case synthesis of two research portfolios, three earlier synthetic tests, and one newly selected source-informed synthetic test; it is not a new general transfer law. The map is an organizing proposal rather than a validated transfer theory.

Results

We examined evidence at distinct levels: connectome-based structure tests (Fig. 3), published biological computations and source-data analyses, and four synthetic experiments on execution-state information, temporal correspondence, fast-slow dynamics and feature-tuned contextual inhibition. NeuroMotif and NeuroMech were independent research portfolios with different tasks and units of analysis; their outcomes were not pooled. M2 and M5 were designed after earlier project outcomes; DMP and the V1 test were later post-result additions. Each protocol was frozen locally before its own outcome run, but none was a project-level preregistration or an independent biological validation (Supplementary Methods S1, S4 and S5; Supplementary Tables S3–S5).

1. The tested connectome descriptions did not establish the proposed computational advantage

The whole-connectome assay did not establish an advantage of the tested off-diagonal topology for value retention (Supplementary Table S1 and Supplementary Results S3). In the frozen reservoir/interference assay, connectome-based interactions did not outperform passive identity-like persistence in the interference range with recoverable signal. A crossover at stronger interference occurred only when both models approached noise-level retention. The comparator retained diagonal state persistence, so the contrast tested the added value of the specified interactions over an existing memory process. It did not compare recurrence with a memoryless system. This result bounded the tested topology, dynamics and task; it did not establish that connectome structure generally lacks computational value.

The separate Fish1.5 analysis asked whether a connectome-derived recurrence measure was positively associated with neuronal activity persistence (Fig. 3a,b). The predicted association was not supported among 82 neurons from one specimen: Spearman's ρ was -0.1096 , the positive-direction permutation P value was 0.8337 , and the neuron-bootstrap 95% confidence interval was $[-0.2932, 0.1338]$. This interval described variation under neuronal resampling within that specimen, not uncertainty across animals. Moreover, the frozen topology-null analysis could not provide valid inference: correlations were undefined for 861 of 1,000 null graphs, leaving 139 defined statistics. No valid topology-null P value was estimable under that procedure. The unsupported positive association and the invalid topology null were therefore retained as separate outcomes; neither justified a population-level structural conclusion. Together, these assays did not establish the proposed structure-to-dynamics advantage under their evaluated conditions.

2. Published source evidence supports bounded computations but does not select an artificial implementation

The motor-state case began with published evidence for a biological computation rather than with an artificial benchmark. Ji et al. reported RIM-dependent motor-state information in AIY and a role for corollary discharge in sustaining forward locomotion during positive thermotaxis in *Caenorhabditis elegans* [3]. This supported a bounded sensory-motor computation at the reported cell-class and behavioural scope. It did not identify the source signal as an exact measurement of displacement after actuator execution, the variable manipulated in the synthetic experiment below.

The project-level animal analysis could not independently resolve that biological account. The available source tables did not provide the animal/session identity and stimulus alignment required for the frozen reanalysis. Consequently, this path remained schema-blocked. Descriptive events could not be treated as independent animals, and the missing alignment prevented the planned test. The blocked result limited the project reanalysis; it supplied no evidence against the published biological computation (Supplementary Table S1).

The cerebellar case also illustrates the distinction between biological observations and an artificial implementation. Garcia-Garcia et al. studied granule-cell and climbing-fibre activity in relation to long-interval timing [4], while optogenetic evidence for climbing-fibre instructive signals in mouse delay eyeblink conditioning supports a biological learning role [5]. Neither source validates the particular artificial update equation tested here. A separate, post-result analysis of released cerebellar data is retained as a branch-pinned exploratory result in Supplementary Results S2; it is not a row in the 39-record Route-D

V3 portfolio and is not included in that portfolio's disposition counts. Together, these source records motivate testing artificial signals and rules separately from biological evidence.

3. Direct execution-state information did not improve the tested action-conditioned controller

We next tested whether an additional state observation improved control in a synthetic actuator-observability problem. A one-dimensional tracking plant combined intermittent, noisy relative-position observations with stochastic actuator reversals. The action-only controller received its preceding command; the realized-state controller additionally received the exact sign of the preceding executed displacement. Both used the same adapted action-conditional recurrent Kalman network (AcRKN) cell architecture, with 320 trainable parameters and equal training budgets [6]. Thirty-two paired training-seed blocks shared environment random streams, with 128 held-out episodes per arm and condition within each block. The independent inferential unit was the training-seed block, not the episode or time step (Supplementary Table S3).

Under hidden reversals, the realized-state observation did not improve the primary outcome. Mean squared tracking error (MSE) was 2.43809 for action-only and 2.50618 for realized-state. The frozen action-only-minus-realized-state contrast was -0.068094 , with a paired seed-block bootstrap 95% confidence interval of $[-0.141077, -0.003541]$. Positive values would have favoured realized-state input; the observed interval instead favoured action-only under this implementation and training budget (Fig. 4b). The input loop is shown in Fig. 4a.

The diagnostic controls did not establish an information-specific explanation for this difference. Supplying the reversal-event bit alongside the command made the realized execution sign reconstructible. In that revealed condition, reversal-event-minus-realized-state MSE was $+0.001992$ (95% confidence interval, $[-0.089678, 0.081652]$). The interval included zero, but no equivalence margin was specified, so this did not establish equivalent performance. The corresponding information-recovery contrast also included zero (-0.070086 ; $[-0.152817, 0.003712]$). With no reversals, when execution sign was determined by the command, the action-only-minus-realized-state gap was -0.041911 ($[-0.101355, 0.021932]$). These diagnostics left input representation and training effects as unresolved explanations; they did not isolate a disadvantage caused by the additional information itself.

Generic recurrent and task-aware references achieved lower mean errors. Under hidden reversals, gated recurrent unit (GRU) controllers reached MSEs of 1.86456 with action-only input and 1.84687 with realized-state input, while the particle-filter reference reached 1.44413. Their mean errors were also lower than the AcRKN arms in the other two conditions (Supplementary Results S4). The GRU pair had 323 parameters each; the particle filter used known task dynamics and was not capacity matched. These comparisons situated AcRKN performance without isolating an architecture effect. Two complete executions reproduced the scientific outputs exactly after exclusion of declared timing fields. The result applied to one plant, an exact uncorrupted post-actuator sign measurement and a fixed training budget. That signal was distinct from the premotor/corollary-discharge signal in the worm source evidence.

4. Preserved temporal correspondence supported prediction without establishing mechanism specificity

The timing experiment tested a different question: whether a fixed local learning rule could exploit trial-specific temporal information when its correspondence with the target was preserved. Each synthetic input contained a noisy temporal pulse related to a target interval. For each of 30 independent task seeds, 1,200 training trials and 400 held-out trials were generated. The aligned regime paired inputs with their generating targets; the broken regime independently permuted target assignments within the training and test splits. The input arrays and target multisets were identical between regimes, so the manipulation changed correspondence while preserving those marginals. The climbing-fibre-inspired local rule used 16 temporal radial-basis eligibility features and a 28-class softmax readout. It was an artificial eligibility-gated update, not a literal model of measured cerebellar long-term depression (Supplementary Table S3).

The task met its frozen viability criterion. In the aligned regime, full-input replay achieved nMAE 0.07138, compared with 0.25816 for the constant-median predictor, a 72.35% relative reduction; its paired correspondence-effect interval was above zero. For the local rule, nMAE rose from 0.141521 when aligned

to 0.258109 when broken (+0.116588; paired task-seed bootstrap 95% CI, [+0.115072, +0.118065]), with a positive contrast in all 30 seeds. Full-input replay changed from 0.071385 to 0.260936 (+0.189552; 95% CI, [+0.186912, +0.192229]), and same-feature ridge from 0.056643 to 0.259033 (+0.202390; 95% CI, [+0.200123, +0.204675]). Both generic effects were larger than the local-rule effect. In contrast, no-trace nMAE changed from 0.258153 to 0.258083 (-0.000070; 95% CI, [-0.000411, +0.000219]) and CF-time-shuffled nMAE from 0.259511 to 0.258312 (-0.001199; 95% CI, [-0.002510, +0.000072]); both intervals crossed zero. Normalization divided interval MAE by the 27-bin target range. Preserved correspondence therefore improved prediction for this fixed local rule, while the generic references used the same task information more effectively (Fig. 5b,c); the algorithm and correspondence manipulation are shown in Fig. 5a.

The temporal controls qualified that effect. Removing the temporal eligibility profile yielded a correspondence-effect estimate of -0.000070, while shuffling teaching-time assignments yielded -0.001199; both bootstrap intervals included zero (Supplementary Results S5). Generic learners exploited the temporal information more effectively: ridge regression on the same eligibility features achieved aligned nMAE of 0.05664 and a correspondence effect of +0.202390 (95% confidence interval, [+0.200123, +0.204675]). Full-input replay showed an effect of +0.189552 ([+0.186912, +0.192229]). The local rule thus demonstrated usable correspondence without outperforming these alternatives.

Comparator matching limited the specificity claim. The ridge arm shared the 16-feature encoder but fitted a 17-parameter scalar head; the local rule fitted a 476-parameter classification head. Their objectives and fitting procedures also differed. The frozen contract's parameter-matched description was corrected in the post-run audit to feature matched, preserving the original record. The comparison showed that a generic learner could use the same representation; it did not isolate a learning-rule effect at equal capacity. In addition, correspondence was changed in both training and evaluation, so its contrast did not separately identify those contributions. The supported result was a correspondence benefit for the tested synthetic implementations, with no established biological mechanism specificity or general transfer advantage.

5. Fast-slow dynamics did not yield a positive mechanism-specific advantage across the tested tasks

Motivated by fast-slow cortical computation, we tested a source-inspired dual-pathway spiking model against a matched short-state control in three synthetic families: delayed association, partially observed state estimation and delayed contextual decision. The model used 32 leaky integrate-and-fire units and a fixed two-dimensional LMU-like state; the control retained the same trainable shapes and state dimension but replaced the slow transition with a fast-decay matrix. A 2×2 design crossed mechanism with whether the slow-context stream remained paired to each trial's fast stream and target. Thirty task-instance seeds per family, two paired initializations and 512 held-out trials per test split were used. The protocol was frozen before this module's run, after earlier project results had been inspected (Supplementary Methods S4).

No task family met the predeclared positive mechanism-specificity criterion, which required a viable task, a positive mechanism $\times$ correspondence interaction after Holm correction and a positive aligned DMP advantage. Delayed association failed its GRU-versus-null viability gate (Holm-adjusted $P = 1.00$); its interaction was -0.00212 (95% CI, -0.00615 to 0.00173; adjusted $P = 0.648$). In state estimation, the interaction was negative (-0.00218; 95% paired task-seed bootstrap CI, -0.00383 to -0.00054), while the aligned DMP model also had higher error than the short-state control; the Holm-adjusted interaction P value was 0.0506. In contextual decision, the interaction was near zero (+0.00101; 95% CI, -0.00537 to 0.00778; adjusted $P = 0.779$), and the aligned full-data advantage was not detected. The study therefore did not show that the tested slow dynamics benefited correctly paired context across these task families.

A predeclared secondary scaling analysis found a narrow small-data pattern in contextual decision: the aligned DMP advantage over the short-state control was +0.0724 at $N = 128$ (95% CI, 0.0504 to 0.0956), +0.0351 at $N = 512$ (0.0134 to 0.0595), and +0.00156 at $N = 2,048$ (-0.00573 to 0.00931). Its unit-level advantage slope was negative (-0.0177 per $\log_2$ training-size step; 95% CI, -0.0229 to -0.0127; Holm-adjusted $P = 0.000060$), meeting the prespecified sample-efficiency gate for this one task. This secondary result did not rescue the failed mechanism-specificity test. In the unchanged longer-delay evaluation,

aligned DMP loss increased by 0.00143 (95% CI, -0.00866 to 0.01081) in delayed association, 0.00096 (-0.00187 to 0.00375) in state estimation, and 0.22096 (0.20153 to 0.23841) in contextual decision. The effect was therefore task dependent, with a substantial degradation in the contextual-decision shift (Supplementary Table S4).

The fast-slow abstraction itself is not new: a dual-memory spiking architecture with compact slow state was reported in *Nature Machine Intelligence* in 2026 [13]. Our implementation was source-inspired, not a reproduction of that model or its hardware co-design. The present synthetic study tests a narrower correspondence interaction and does not establish cortical mechanism validity, energy efficiency, broad transfer or novelty of fast-slow memory.

6. Feature-matched inhibition did not improve the synthetic figure-ground task

We tested a reported mouse V1 motif—like-to-like orientation-tuned pyramidal-cell input to SST interneurons and SST contribution to surround suppression [14]—on a synthetic figure/ground task. Generated eight-channel orientation responses, not source neural recordings, were used to segment a 3×3 patch. Thirty task-instance seeds crossed selective versus global inhibition with aligned versus permuted routing; training sizes were 32, 128 and 512 images, with IID and skewed-background tests (Supplementary Methods S5 and Table S5). At $N = 512$, the generic-context reference exceeded the 9/121 AP null by 0.37219 (95% task-seed bootstrap CI, 0.36046 to 0.38379), passing task viability.

The frozen interaction was +0.01603 AP (95% CI, 0.00815 to 0.02357; two-sided sign-flip $P = 0.00060$). Its formula was broken minus aligned, so with higher-is-better AP, the estimate indicates a relative benefit from breaking routing—the opposite of the protocol narrative's prediction. Selective/aligned pooling also underperformed global/aligned pooling by 0.02208 AP (95% CI, 0.01664 to 0.02737). The positive mechanism-specificity gate failed. The skewed-background interaction had the same direction (+0.01692; 95% CI, 0.00553 to 0.02913); the sample-size slope was inconclusive. We preserved the frozen contract and outputs and documented its sign-direction wording error.

This source-informed synthetic test was selected after earlier project outcomes; its visual domain overlaps prior synthetic-visual work. Its generic comparator was a linear context model, not a strong or parameter-matched neural network. It does not validate the biological circuit, the full evidence map or a general transfer claim.

Discussion

The evaluated cases separated source support from artificial utility at several points. Structural descriptions did not establish their proposed dynamical advantage within the tested assays. Supported biological computations left artificial implementation choices unresolved. The synthetic experiments had different outcomes: direct execution-state access did not improve the tested AcRKN controller; temporal correspondence helped a local rule but generic alternatives had larger effects; fast-slow dynamics did not yield a positive mechanism-specific advantage across three task families; and feature-selective inhibition failed its aligned-utility gate in the new segmentation task. The latter interaction ran opposite to its stated directional prediction, so it does not support an alignment benefit. Together, the results treat source support, information correspondence and implementation utility as separate, evidence-bounded claims.

We propose a neural transfer evidence map with four questions: source validity, computational interpretation, information correspondence and artificial utility. Source validity concerns the identity, coverage and limitations of the biological observations. Computational interpretation specifies the operation supported by recordings, models or interventions. Information correspondence asks which task-relevant variable or relation survives in the artificial input and representation. Artificial utility is a measured benefit under a stated objective and comparator set. Mechanism specificity is a separate question about whether the proposed operation explains an effect beyond alternative implementations. The map organizes the present cases; its categories are neither a causal sequence nor established necessary and sufficient conditions for successful transfer (Fig. 2).

The synthetic outcomes are compatible with implementation-level explanations. In M2, worse performance from additional information does not imply that the observation is intrinsically harmful or useless to an optimal estimator; diagnostics left representation and training effects unresolved. In M5, ridge used the same eligibility features but a smaller head and different objective, so its advantage leaves the learning-rule contribution unidentified at matched capacity. The correspondence manipulation changed both training and evaluation pairing, and complete-trial features did not demonstrate online eligibility. In the fast-slow study, the short-state control performed better in state estimation; contextual-decision gains diminished with data and did not persist at longer delay. This may reflect task-specific optimization, not a biological cause.

These findings qualify the evaluated implementations without overturning successful connectome-informed or biologically motivated models in other settings. Connectome reservoir models have shown task-specific utility, while recent work on connectome-constrained recurrent networks combines structural and functional evidence to predict activity [1, 2]. These precedents place the present negative or bounded results in context: they do not show that biological connectivity lacks computational value. Source circuitry can constrain a model without being the uniquely effective artificial realization of a computation. Conversely, generic prediction from a source-derived representation does not show that the source mechanism is absent or unnecessary in the animal. The evidence map makes those distinctions explicit (Fig. 6), but these cases do not establish its novelty or generality across neural systems.

Several limits follow from the evidence units and selection history. Fish1.5 included one specimen and an invalid frozen topology null; neuronal resampling did not provide animal-population inference. The source readout used within-session splits and modeled weights (Supplementary Results S2). All four synthetic tests followed inspection of earlier outcomes. The new V1 test used generated visual signals, a linear generic comparator and a contrast with a sign-direction wording error; its modality also overlaps earlier project work. DMP had no biological measurements or independent transfer, and its scaling signal arose in one family without specificity. Historical invalid, blocked and unfavourable records remain in the supplement (Supplementary Results S1). Source linkage and reuse rights remain incomplete. These cases support bounded implementation claims only.

Methods

Study design and evidence selection

NeuroConverge is a retrospective synthesis of two independently developed research portfolios, with four separately analysed synthetic tests. It is not a systematic review, a prospectively unified biological study or a meta-analysis. We imported all 39 records from the NeuroMech Route-D V3 classification matrix and retained their original evidence labels, units, statuses and interpretation limits in Supplementary Table S1. Selected packages on the separate experiment-publication branch were crosswalked to that matrix (Supplementary Table S2). Some packages lack an exact matrix record; this limits the completeness of the source inventory, not the identities of the separately hashed packages. File identities and source states are summarized in Supplementary Methods S2. RR18 and RR19 belong to separately governed workstreams and were excluded from this synthesis.

Main-text selection considered structural evidence, source-supported computation, execution-state access, temporal correspondence and fast-slow dynamics. Selection was made with knowledge of earlier project outcomes. Historical positive, negative, invalid and inconclusive records were retained in the supplementary portfolio. An invalid analysis or failed viability gate was not interpreted as a valid negative mechanism test. Neither the portfolios nor the M2, M5 and DMP V1 endpoints were pooled. Each module contract was frozen before its own outcome execution, following inspection of earlier project results; this was not project-level preregistration, and none is an independent biological confirmation.

Whole-connectome reservoir assay

The locally archived M0 report evaluated neuron-level mappings of worm and larval fly connectivity under fixed reservoir dynamics and sustained interference. The recurrence implemented

$$h_t = (1 - \lambda)h_{t-1} + \lambda \tanh(W_{hh}h_{t-1} + W_{hx}x_t + b).$$

The passive comparator set the off-diagonal recurrent matrix to zero while retaining the same leak and diagonal state persistence. Matched comparisons shared the input-weight draw, stimuli, interference realization, probe split and ridge regularization. Value retention was assessed using readouts of reservoir state under the predefined delays and interference settings. The manuscript uses the corrected report's bounded conclusion in the range with recoverable information, retaining its noise-floor crossover qualification. The underlying report also records node-exclusion, random-seed and probe corrections; earlier invalid implementations were not treated as independent results.

This manuscript did not rerun or reconstruct the complete M0 assay. The corrected report and available manifests are identified in the paper-critical source snapshot. Canonical round-level linkage is not fully reconciled, so the present Methods describe the supported comparator definition without assigning one unresolved configuration to every historical round. The report, scope of this limitation and source records are retained in Supplementary Results S3 and the reproducibility inventory.

Fish1.5 structure and functional persistence

The Fish1.5 analysis used released structural annotations from the version-2 preprint source [7] and archived traced axons/synapse tables (Zenodo record 19231045; [9]), together with functional calcium-response arrays from one specimen (Zenodo record 16893093; [8]). Of 97 structurally covered functional identities, the preselected primary cohort comprised 82 with strict structural completeness; the expanded 97-neuron graph was sensitivity-only. Graph construction used each mapped neuron's presynaptic owner table. Only annotations marked valid and partners mapped into the chosen cohort were retained. Self edges were excluded. Repeated synapse identifiers were collapsed only when partner coordinates agreed; ambiguous duplicates were rejected. Directed weights counted retained annotations rather than missing size fields. Postsynaptic files were integrity checks, not sources for filling absent primary edges.

For adjacency matrix W , where W_{ij} is the directed weight from neuron i to neuron j , define out-strength $_i = \sum_j W_{ij}$ and in-strength $_i = \sum_j W_{ji}$. The local recurrence index was

$$L_i = \sum_j W_{ij} W_{ji} / [(out-strength_i)(in-strength_i)],$$

with zero assigned when the denominator was zero. The numerator is the diagonal of W^2 , not of WW^T . The functional endpoint used nominal timing defined by the released analysis code: 160 samples at 0.5-s spacing and dot stimulation during [20,60) s. Baseline, late-stimulus and post-offset windows were [0,20), [50,60) and [60,70) s. A trial required finite values in all three windows. Its persistence ratio was the baseline-subtracted mean post-offset response divided by the baseline-subtracted late-stimulus response. Exactly zero or nonfinite denominators were undefined and recorded, without an amplitude threshold or imputation. Direction-specific trial ratios were summarized by medians; the available left/right direction summaries were averaged per neuron.

The primary statistic was Spearman correlation between L_i and persistence among eligible primary neurons. A positive-direction permutation test shuffled persistence labels 10,000 times and used the add-one upper-tail fraction. A 10,000-draw paired neuron bootstrap produced a percentile interval from defined correlations. The valid draw counts and the descriptive two-sided permutation result were retained in the source outputs; the bootstrap interval was conditional on the specimen and did not capture all network dependence or animal variation. Random streams were derived from seed 2026093001.

The frozen topology null generated 1,000 unique directed binary graphs by degree-preserving double-edge swaps, retaining the binary in/out degree sequences and global positive-weight multiset. Positive weights were randomly assigned to rewired edges; node-specific weighted strengths were not preserved. Every null correlation had to be defined for the frozen topology comparison to be valid. Undefined statistics were retained and the topology P value was left unestimated. Fixed secondary descriptors, Holm-adjusted secondary associations and robustness analyses remained distinct from the primary result. Phase lag and calibrated sine gain were not identifiable from the released schema.

Biological source evidence and descriptive reanalyses

Published findings were attributed to their primary studies and were not represented as experiments performed for NeuroConverge. The worm source concerned RIM-dependent motor-state information in AIY and thermotaxis [3]. The planned animal-level reanalysis required identities and stimulus alignment unavailable in the relevant released tables. A separate project event-sheet summary retained event-weighted descriptions only; events were not used as animal replicates. These paths were kept distinct from the generic M2 synthetic plant.

The separate cerebellar source-readout analysis and its complete procedure are documented in Supplementary Methods S3 and Supplementary Results S2. This post-result package is pinned to a divergent publication branch and is presented as exploratory supplementary evidence, not as one of the 39 Route-D V3 records or part of their disposition counts.

M2 V1 plant and causal inputs

The input, controller, actuator and feedback sequence is summarized in Fig. 4a. M2 V1 used a 64-step differentiable one-dimensional tracking plant. Target position q_0 was uniform on $[-1, 1]$, initial target velocity was Gaussian with SD 0.055, and subsequent velocity followed $v_t = 0.91v_{t-1} + \varepsilon_t$, with independent Gaussian innovations of SD 0.055; position was updated by velocity. Agent position began at zero and changed after a bounded command $a_t \in [-1, 1]$ by $p_{t+1} = p_t + 0.40a_t s_t$, where $s_t = -1$ indicated actuator reversal. Training reversal probabilities were sampled independently per episode uniformly from $[0, 0.25]$. Held-out conditions used probability zero or 0.25; hidden and revealed reversals shared trajectories of environmental random variables.

Relative-position observations had Gaussian noise SD 0.14. Missingness used a two-state Markov process initialized with missing probability 0.5 and transition probabilities 0.125 into and out of the missing state, giving mean missing-burst duration eight steps. Six input features were supplied in fixed order: observed relative position or zero, visual-valid flag, previous execution auxiliary value, auxiliary-valid flag, previous command and visual gap clipped at 16 steps and divided by 16. The auxiliary value was zero with validity zero for action-only; the sign of previous executed displacement for realized-state; or +1 for a previous reversal and -1 otherwise for the revealed-event arm. Auxiliary validity began at zero. The revealed-event arm received its event channel during training and the no/revealed-reversal conditions, but not the hidden condition. None received a future or current unexecuted reversal event.

M2 V1 models, training and reference

The adapted AcRKN cell used a six-to-three linear observation encoder, a six-dimensional latent mean and structured covariance, one transition basis, bandwidth two and a 26-unit action-control subnetwork. The policy head mapped the posterior mean through three tanh units to a tanh scalar command. Initial state and observation-variance parameters were learned. Mathematically inert one-basis coefficient-network parameters were frozen and excluded from the trainable count. Encoded observations were treated as valid at each cell update; modality missingness was represented by input flags. The observation update and policy preceded the action-conditioned prediction for the next step. The vendored cell retained its training-time dropout. This wrapper was an adaptation, not a complete reproduction of the original robotics model.

The three AcRKN arms each had 320 trainable parameters. The two generic GRU arms used the same six inputs, eight hidden units and a scalar policy readout, with 323 parameters each. Within family, initialization and environmental streams were paired. Each learned model used 160 Adam updates at learning rate 0.003, batch size 32, full 64-step sequences and global gradient clipping at five. The loss averaged squared target-relative distance measured before each step's new displacement. Gradients propagated through the differentiable rollout; the executed-displacement sign observation was detached. Each fit consumed 327,680 training time steps. Training and test streams were separately identified in the retained registry.

The task-aware particle filter used known target, observation and actuation distributions. Its 256-particle main evaluation inferred reversals when hidden and received preceding reversal events when revealed. Observation likelihoods updated weights, and systematic resampling occurred below half the particle count

in effective sample size. The command was $\tanh(k_p \hat{e} - k_d \hat{v}_{\text{agent}})$. Gains were selected from $k_p \in \{0.45, 0.70, 0.95, 1.20\}$ and $k_d \in \{0, 0.15, 0.30\}$ on four disjoint validation seeds with 32 episodes each, using 64 particles and hidden reversals. The selected gains were held fixed in test evaluation. Particle counts 64, 128 and 256 and effective-sample-size diagnostics were retained. The reference was not matched in information, capacity or inference compute to learned models.

M2 V1 endpoint and reproducibility

The independent unit was one paired training-seed block, with seeds 950000–950031. Each arm and condition used 128 test episodes per block. Episode-mean tracking MSE was averaged within block. The primary contrast was action-only minus realized-state AcRKN MSE under hidden reversals. Diagnostic contrasts comprised revealed-event minus realized-state in the revealed condition, the primary minus that revealed gap, and action-only minus realized-state under no reversals. Percentile intervals resampled paired block contrasts 20,000 times using seed 7813921. They were unadjusted exploratory intervals; no equivalence margin or portfolio-wide error control was specified.

Two full runs used Python 3.9.6, NumPy 2.0.2 and PyTorch 2.8.0 on CPU with one thread. Stored-output verifiers checked summaries, pairing, causal feature tests and hashes. Scientific files were compared byte for byte, excluding only four declared timing fields. Complete fit records, model parameters, episode metrics, stream identifiers, parameter counts and particle diagnostics were retained. Base source commit identity did not substitute for the captured contract, runner and verifier hashes in SOURCE_PROVENANCE.json.

M5 V1 generator, features and learning

The synthetic input, offline feature construction, local update and correspondence manipulation are summarized in Fig. 5a. M5 V1 used task seeds 51000–51029. Each seed independently generated 1,200 training and 400 test trials with 40 bins ($t = 0, \dots, 39$). Targets y were uniform integers 6–33. Inputs were $x_t = \exp[-(t-c)^2 / (2 \times 2.25^2)] + \varepsilon_t$, with $c = y + \xi$, Gaussian ξ SD 1.5 and independent Gaussian observation noise SD 0.20. Broken correspondence independently permuted training and test targets, preserving complete input arrays and target multisets. Permutations had no fixed trial indices; after at most 100 failed random draws, a cyclic shift was used. Repeated target values could still yield coincident labels after permutation. A separate permutation defined the shuffled-teaching arm.

The eligibility representation projected positive-clipped complete inputs onto 16 Gaussian bases with evenly spaced centres from 0 to 39 and width 2.75. Feature vectors were normalized by their Euclidean norm, with a 10^{-12} floor, then augmented by a bias. This calculation consumed the full trial array; it was not implemented online within-trial trace. The local rule processed generated training trials once in their existing order. With zero-initialized weights, class probabilities $r = \text{softmax}(\tilde{\phi}^T W)$, and a one-hot target z_v , the update was $W \leftarrow W + 0.08 \tilde{\phi}(z_v - r)^T$. Predictions were probability-weighted target-bin means. The teaching-time shuffle changed training labels for this update; evaluation retained each regime's actual targets. The no-trace arm used final-bin input and trial mean with bias under the same softmax rule, so it removed the temporal profile rather than every historical summary.

Generic ridge used the same 16 features and a scalar head. Full-input replay used all 40 input bins with a scalar head. Both fitted β by solving $((A^T A + 0.01 D) \beta = A^T y)$, where the bias penalty in diagonal (D) was zero, and clipped predictions to $[6, 33]$. The generic ridge had 17 fitted output parameters; local/shuffled rules had 476, no-trace had 84 and full-input replay had 41. These arms differed in objective and fitting procedure as well as head capacity. The frozen contract's parameter-matched label was corrected after the run to feature matched without altering recorded outcomes.

M5 V1 endpoint, gate and execution record

The sole performance metric was test MAE divided by 27 bins. The independent unit was task seed. The primary contrast was local-rule nMAE(broken) minus nMAE(aligned); comparator contrasts were descriptive secondary results. A deterministic 20,000-draw paired-seed percentile bootstrap used a generator initialized with seed 2026100105 and advanced across arms in the recorded order. These intervals were not multiplicity adjusted. Task viability required at least 25% relative mean improvement of

aligned full-input replay over a training-target median predictor and a strictly positive lower interval bound for replay's correspondence effect. Conditional on viability, a positive primary lower bound supported the frozen correspondence claim; an interval spanning zero was classified as not detected, and an entirely negative interval as reversed.

The first launch failed before producing outcomes because Python 3.9 did not support two zip(strict=True) calls. A syntax-only amendment was frozen before the one successful run under Python 3.9.6 and NumPy 2.0.2. Original and amended freezes and execution-incident records were preserved. Reported NumPy/Accelerate warnings were audited against finite outputs and an independent contraction check; the independent result verifier passed. No arm, seed, model or generator was changed after outcome inspection. The manifest identifies the successful runner, contract and all scientific output hashes. The original labels in frozen files were preserved as historical provenance, even where the manuscript uses corrected terminology.

DMP fast-slow correspondence V1 tasks, models and inference

This synthetic post-result extension was inspired by the dual-memory spiking abstraction reported by Sun et al. [13]. It was not a reproduction of their network or hardware. Three task generators were implemented: (1) a noisy four-level context at steps 0–3 and a four-level query at steps 36–39, mapped to four classes; (2) a unit-variance AR(1) latent state observed through two channels whose relative noise levels were indicated by a switching context, with the final latent value as target; and (3) an early four-level context selecting a mapping from four noisy evidence votes at steps 36–51 to one of four actions. Each task instance also generated an unchanged longer-delay test: query/evidence moved to steps 68–71/68–83 in a 128-step sequence, or state-estimation horizon doubled from 64 to 128. Configuration parameters and class maps were fixed per task-instance seed.

The DMP-style model had 32 leaky integrate-and-fire units (leak 0.5, threshold 1, subtractive reset) and a fixed two-dimensional Legendre-like slow state discretized at horizon 32. The short-state control retained the same learnable parameters, input matrix and state dimension, but used a 0.5I transition. A 2×2 factorial crossed model with aligned or broken slow-context correspondence. Broken correspondence independently deranged the context streams within each training prefix ($N = 128, 512$ or $2,048$) and each held-out split, preserving exact context marginals and leaving fast streams, targets and counts unchanged. The comparator GRU had 18 hidden units and trainable parameter count within 5% of DMP. All learned models used two paired initializations, ten epochs, batch size 64 and Adam at learning rate 0.003. FLOPs were not matched.

Thirty independent task-instance seeds were run per family. Each generated 2,048 training trials, 512 IID test trials and 512 longer-delay trials. Classification used 0–1 loss; state estimation used squared error against the unit-variance latent target. The primary family-specific contrast was the mechanism $\times$ correspondence interaction, $I = [L(\text{DMP}, \text{broken}) - L(\text{DMP}, \text{aligned})] - [L(\text{control}, \text{broken}) - L(\text{control}, \text{aligned})]$. The task gate required the aligned full-data GRU to beat the task-null predictor. Paired task-seed percentile bootstrap intervals and two-sided sign-flip tests were calculated at the task-instance level; Holm correction covered the three co-primary interactions and separately the three viability tests. The scaling gate required a positive 95% interval for the aligned DMP advantage at $N = 128$, negative advantage slope versus $\log_2(N)$, and Holm-adjusted slope $P < 0.05$. Longer-delay results were secondary. The pre-run freeze, source/runner/verifier hashes, per-trial losses and independent verification record are retained in `../new_experiments/DMP_FAST_SLOW_CORRESPONDENCE_2X2_V1/`.

Prospective V1 PC→SST feature-correspondence benchmark

This source-informed synthetic package was motivated by Hendricks et al. [14], who reported like-to-like functional connectivity from V1 pyramidal cells to SST interneurons and causal SST contribution to orientation-dependent surround suppression. We did not download or analyse the paper's Zenodo recordings. Each synthetic image comprised an 11×11 grid and eight orientation channels spaced over 180° . Responses were generated from a fixed soft orientation-tuning function ($\kappa = 2$) with Gaussian noise (s.d. = 0.08), clipped positive and normalized across channels. One randomly placed 3×3 figure had an orientation sampled uniformly from the seven categories other than its uniform-background category. Each

task-instance seed generated 512 training images, 256 IID test images and 256 test images with a fixed skewed background-orientation prior; the same nested training prefixes of $N = 32, 128$ and 512 were used across arms.

For each location, the local surround was the mean of its eight neighbouring responses. The feature-selective pathway subtracted 0.65 times the same-channel surround activity, or the activity routed through a fixed no-fixed-point channel permutation in the correspondence-broken arm. The global-pool control subtracted 0.65 times the mean surround activity, broadcast across channels, and was invariant to that permutation. All four factorial cells used the same ridge-regularized linear discriminant readout ($\lambda = 0.02$); the generic-context reference received concatenated local and surround channels (16 inputs). This generic reference was information matched but neither parameter matched to the eight-input factorial cells nor a strong neural network. The task-null AP was the positive-pixel prevalence, 9/121.

Thirty task-instance seeds were the independent units; images and pixels were nested. The primary outcome was mean per-image AP. The frozen interaction used the (broken – aligned) AP contrast for selective minus global pooling; the protocol narrative mistakenly assigned a positive value to an alignment benefit. We preserve that formula and report the observed sign with its corrected directional meaning. Paired task-seed bootstrap intervals used 20,000 draws; two-sided sign-flip tests used 100,000 Monte Carlo sign patterns with NumPy default_rng(20261003). The positive specificity gate required generic-reference viability, an interaction meeting the frozen numerical criterion, and higher aligned AP for selective than global pooling. The three sample sizes and the skewed-background test were secondary. The full protocol, source/runner/verifier hashes, per-image outcomes, verifier and amendment are in `../new_experiments/PROSPECTIVE_V1/`.

Units, inference and writing assistance

Neurons within one specimen, source recording sessions, synthetic training-seed blocks and synthetic task seeds were kept distinct. Trial, episode, frame, null-graph and bootstrap counts were not described as biological replicate counts. No new animal experiment was conducted for this integration. Source-study ethical approvals remain those reported by the original authors; no approval identifier was invented or transferred to the present synthetic work. Published biological conclusions were not reclassified based on synthetic benchmark outcomes.

AI assistance was used to organize the evidence inventory, draft and revise manuscript prose, inspect code and source records, and generate internal-review figure layouts. Numerical reporting was reconciled with retained outputs and reviewed separately. This statement describes the assistance performed; final scientific interpretation, attribution and submission remain the responsibility of the human authors.

Figure legends

Figure 1 | Why biological provenance does not specify artificial utility. Source anatomy, activity, perturbation and behaviour can motivate a computation, but translation still requires choices about signal, representation, objective and learning procedure. Utility is evaluated against a task and comparator set. The diagram motivates the evidence questions; it is not a causal model or an empirical result.

Figure 2 | Retrospective evidence framework for the NeuroMotif and NeuroMech portfolios. Four columns distinguish biological structure, source-supported computation, information correspondence and artificial utility. Colored bars indicate which portions of this retrospective portfolio each project addressed; dotted separators mean each transition requires a separate test. The arrangement is a proposed evidence map, not a causal chain, validated hierarchy or law.

Figure 3 | Fish1.5 structure-to-dynamics evidence. (a) Recurrence–persistence Spearman correlation in one specimen and 82 structurally complete neurons ($\rho = -0.110$; 95% neuron-bootstrap CI, -0.293 to 0.134 ; positive-direction permutation $P = 0.834$). (b) Of 1,000 frozen topology-null graphs, 139 yielded defined correlations and 861 did not; the topology-null P value was not estimable. Neuron-level resampling does not represent between-animal uncertainty.

Figure 4 | Synthetic actuator-observability test. (a) The two AcRKN arms differ in whether the controller receives the exact sign of the preceding executed displacement. (b) Paired seed-block mean-MSE contrasts (first arm minus second; positive values favour the second, lower-MSE arm; $n = 32$). Points show means and bars show paired seed-block bootstrap 95% confidence intervals. Only the hidden-reversal primary contrast excluded zero, favouring action-only input. This synthetic post-actuator signal is distinct from worm premotor RIM–AIY signalling; the test was outcome-informed.

Figure 5 | M5 synthetic temporal-correspondence test. (a) Input, eligibility features, local update and aligned or broken target correspondence. (b) Paired task-seed nMAE effects (broken minus aligned; positive values favour preserved correspondence; $n = 30$). Points show means and bars show paired task-seed bootstrap 95% confidence intervals. (c) Mean nMAE by regime (lower is better); row order matches panel b. Ridge shares features with the local rule but differs in head, objective and fitting; replay receives all input bins. The result is task-specific and does not establish mechanism specificity.

Figure 6 | Evidence status and inference ceilings. Each row identifies a distinct evidence question, its record-level status and narrowest supported scope. The figure includes the DMP fast-slow study and the source-informed V1 PC–SST synthetic test. M2, M5, DMP and V1 endpoints and independent units are not combined. The V1 interaction was positive under a broken-minus-aligned AP formula, opposite to the protocol narrative; aligned selective pooling also underperformed the global-pool control. This test is synthetic, selected after earlier project outcomes, and shares the broader visual domain with prior work. “Not established” denotes a limit of assessed evidence, not proof of absence. The map remains a retrospective organizing proposal, not a validated hierarchy.

Data availability

This manuscript uses published or released datasets and source files identified in the Methods and Supplementary Information. The Fish1.5 functional and structural datasets are available through Zenodo (DOIs: 10.5281/zenodo.16893093 and 10.5281/zenodo.19231045; both CC BY 4.0); the cerebellar source-readout data are available through Dryad (DOI: 10.5061/dryad.bk3j9kdm6; CC0 1.0). The source V1 data/code archive is deposited at Zenodo (10.5281/zenodo.17527499), but it was not used as an observation set in the synthetic benchmark. The M2, M5, DMP and prospective V1 synthetic-run outputs, contracts, verifiers and hashes are included in this project workspace. Public source linkage and redistribution rights for all imported records are still being audited; repository contents should not be assumed to confer third-party redistribution rights.

Code availability

The M2, M5, DMP and prospective V1 generators, runners, frozen contracts, verification code and source-linked figure-generation script are included in this repository. Source hashes and execution details are recorded in the corresponding run manifests and Supplementary Information. The V1 post-run amendment records a directional wording error without changing its frozen contract or results. Public release of a final archival code snapshot remains pending the source-freeze and licensing audit.

Author information and declarations

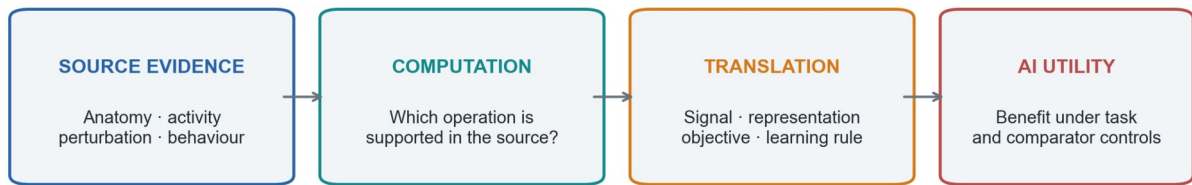
Authors and affiliations: to be supplied and approved by the human authors. **Author contributions:** to be assigned by the authors. **Funding:** to be supplied by the authors. **Competing interests:** to be declared by the authors. These details were not available in the project record and have not been inferred.

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Figures

Biological evidence leaves key implementation choices open



Each transition needs its own evidence and comparator

A biological label alone does not fix the model input, objective, training procedure or the alternative that defines an advantage.

Motivating distinction; not a causal model or a result established by these portfolios.

Figure 1 | Why biological provenance does not specify artificial utility. Source anatomy, activity, perturbation and behaviour can motivate a computation, but translation still requires choices about signal, representation, objective and learning procedure. Utility is evaluated against a task and comparator set. The diagram motivates the evidence questions; it is not a causal model or an empirical result.

Two evidence portfolios ask four distinct questions

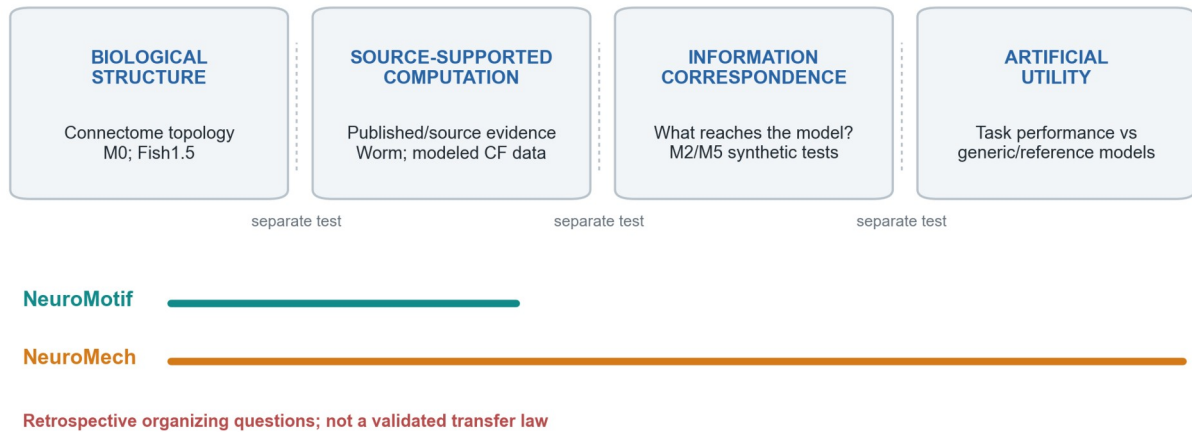
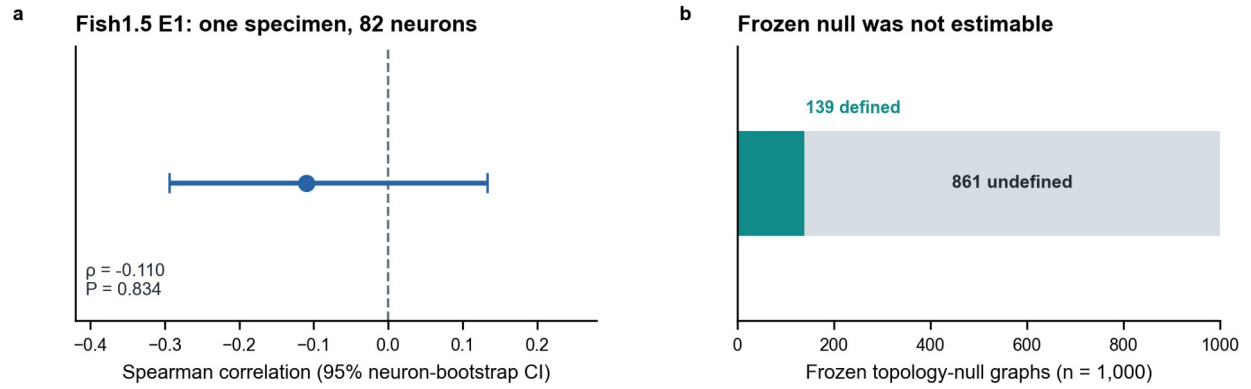
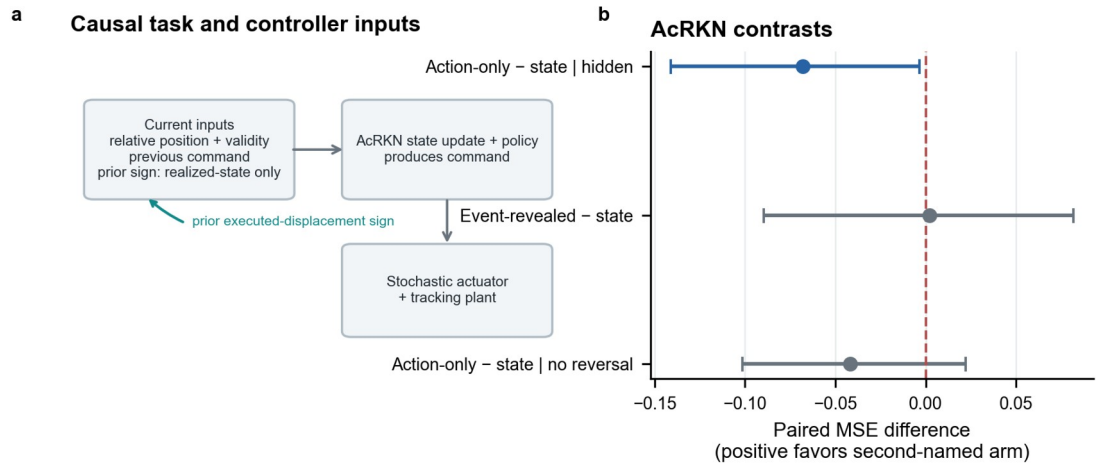


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Fish1.5 structure-to-dynamics evidence is indeterminate

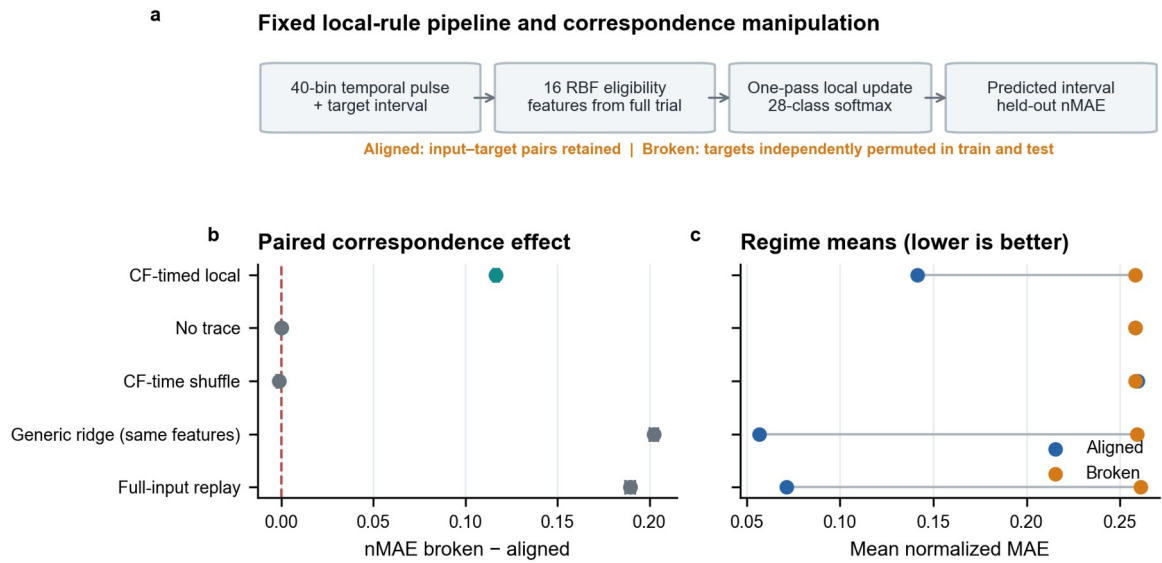
One specimen; neuron-level resampling does not estimate between-animal uncertainty.
The frozen topology-null P value was not estimable because most null correlations were undefined.

Figure 3 | Fish1.5 structure-to-dynamics evidence. (a) Recurrence–persistence Spearman correlation in one specimen and 82 structurally complete neurons ($\rho = -0.110$; 95% neuron-bootstrap CI, -0.293 to 0.134 ; positive-direction permutation $P = 0.834$). (b) Of 1,000 frozen topology-null graphs, 139 yielded defined correlations and 861 did not; the topology-null P value was not estimable. Neuron-level resampling does not represent between-animal uncertainty.

M2: exact execution feedback did not improve the tested controller

32 paired training-seed blocks; error bars are paired block-bootstrap 95% CIs. The primary hidden-reversal contrast favored action-only. Synthetic post-actuator sign is not the worm premotor RIM→AIY signal; outcome-informed test, not biological-transfer validation.

Figure 4 | Synthetic actuator-observability test. (a) The two AcRKN arms differ in whether the controller receives the exact sign of the preceding executed displacement. (b) Paired seed-block mean-MSE contrasts (first arm minus second; positive values favour the second, lower-MSE arm; $n = 32$). Points show means and bars show paired seed-block bootstrap 95% confidence intervals. Only the hidden-reversal primary contrast excluded zero, favouring action-only input. This synthetic post-actuator signal is distinct from worm premotor RIM→AIY signalling; the test was outcome-informed.

M5: preserved correspondence helped prediction, not mechanism specificity

30 paired task seeds; paired-seed bootstrap 95% CIs. Outcome-informed test; full-trial features are offline. Ridge shares 16 features but has a different, smaller head and objective; replay receives all 40 bins.

Figure 5 | M5 synthetic temporal-correspondence test. (a) Input, eligibility features, local update and aligned or broken target correspondence. (b) Paired task-seed nMAE effects (broken minus aligned; positive values favour preserved correspondence; $n = 30$). Points show means and bars show paired task-seed bootstrap 95% confidence intervals. (c) Mean nMAE by regime (lower is better); row order matches panel b. Ridge shares features with the local rule but differs in head, objective and fitting; replay receives all input bins. The result is task-specific and does not establish mechanism specificity.

Evidence map: what is supported, blocked or unresolved

Evidence question	Record-level status	Inference ceiling
Connectome structure → task computation	M0 bounded (round linkage pending); Fish1.5 association unsupported; null indeterminate	Frozen assay / one specimen
Biological computation in source system	Published worm evidence; project reanalysis schema-blocked	Published cell-class scope
Execution feedback → artificial utility	M2: realized-state input worse than action-only; GRU/PF lower MSE	One synthetic plant
Temporal correspondence → artificial utility	M5: local-rule effect; generic ridge and replay effects larger	One synthetic generator
Fast–slow dynamics × correspondence	DMP: no family passed specificity; contextual small-data signal only	Three synthetic families; post-result
Feature-matched inhibition × correspondence	V1: interaction opposite prediction; aligned selective pool worse	One synthetic visual task; sign error
General transfer law	Not established by portfolios or V1 test	Retrospective synthesis + one synthetic test

Endpoints and units differ; no cross-task effect is calculated. V1 was selected after earlier results, overlaps the prior visual domain, and its frozen AP contrast had a sign-direction wording error. This evidence map is retrospective, not a validated causal hierarchy.

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